MotorSentinel-RV System Architecture

Track 2 — PolarFire SoC Icicle Kit | sensor-bus guard + sensor-integrity monitor + INT8 EdgeMatrix-RV autoencoder + hardware safety policy

ADXL345
3-axis accelerometer
800 S/s, 256 LSB/g

I2C 400 kHz

RP2040 PIO
attack emulator
replay, freeze, NACK, bias

PolarFire SoC MPFS250T

FPGA FABRIC — deterministic data plane (motorsentinel_core_apb)

I2C Acquisition
ADXL345 sequencer
(next milestone)

X/Y/Z

SCL/SDA

Sensor-Integrity Monitor
range + slew limits
frozen-value run detector
replay: 16-tap rolling hash
1024-entry tag RAM

range / slew / frozen / replay

Model Store
122-word image, CRC-32
format + range checks
fail-safe atomic commit

weights

Protocol Guard
2-FF line sync
stuck SCL/SDA, stretch
NACK storm, rogue START
missing-sample watchdog

Feature Extractor
256-sample / 128-hop windows
mean, MS, p-p, exact ZCR
4 Haar band energies
16 x 32-bit, bit-exact

INT8 Normalizer
per-feature offset,
scale, shift, sat8
streams with the
16-word transfer

q[16]

EdgeMatrix-RV
8x8 INT8 systolic array
16 -> 8 -> 16 autoencoder
4 weight tiles, ReLU
L1 score in 74 cycles

score

Safety Policy
verified kernel, unchanged

protocol faults
integrity faults
learned anomaly
(2-window confirm)
latched provenance
healthy-window recovery
explicit re-arm

protocol faults (6 causes)

APB3 via FIC3, one 256-byte slot: 0x00-0x3F safety kernel, 0x40-0xFF data plane | level IRQ to MSS

RISC-V MSS — Linux control plane (not in the safety path)

UIO driver + model loader
CRC-checked image, thresholds

Event service + dashboard
fault provenance, scores, features

CPU baseline + replay
same data via APB injection

PAC1934 power monitor
CPU vs FPGA energy per window

safe disable

Motor Driver
+ DC motor
active-high enable,
pull-down fail-safe

Verified in RTL simulation

- 9/9 attack scenarios pass
- every score bit-exact vs Python
- decision 348 cycles (6.96 us)
- frozen sensor trip: 38.75 ms
- spike trip: 1 sample (1.25 ms)
- mechanical faults: 320 ms

implemented and verified
planned / board integration
independent hardware safety response
sensor-integrity evidence